

Fisher Information and Stochastic Complexity

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Abstract—By taking into account the Fisher information and removing an inherent redundancy in earlier two-part codes, a sharper code length as the stochastic complexity and the associated universal process are derived for a class of parametric processes. The main condition required is that the maximum-likelihood estimates satisfy the Central Limit Theorem. The same code length is also obtained from the so-called maximum-likelihood code.

Index Terms—Universal coding, universal modeling, MDL principle.

I. INTRODUCTION

THE idea of universal coding, suggested by Kolmogorov, is to construct a code for data sequences such that asymptotically, as the length of the sequence increases, the mean per symbol code length would approach the entropy of whatever process in a family has generated the data. In the seminal works by Davisson [5], [6], as well as by Krichevsky and Trofimov [9], Rissanen [13], Shtarkov [19], and others, this has been shown to be possible for many of the usual types of finite alphabet processes. Different universal codes can be compared in terms of the mean code redundancy of the worst case process, i.e., the mean code length excess over the entropy, maximized over the processes in the considered family. An interesting result due to Gallager in 1974 (unpublished lecture notes) states that the worst case mean redundancy for the best code equals the channel capacity, when the family of processes $\{f(x^n | \theta)\}$, together with a prior $w(\theta)$, is viewed as an information channel. The capacity, then, is defined as the maximized mutual information $I_w(\Theta; X) = H_w(\Theta) - H_w(\Theta | X)$ over the priors. The same fact was found later for Markov processes by Davisson and Leon-Garcia in [7]; see also [18]. Such a maximizing prior $\pi(\theta)$ was defined by Bernardo in [2] with an imaginative Bayesian type of interpretation as the *reference prior*. It coincides in the scalar cases with Jeffreys' prior

$$\pi(\theta) = \frac{|I(\theta)|^{1/2}}{\int |I(\eta)|^{1/2} d\eta} \quad (1)$$

where $I(\theta)$ is the Fisher information matrix

$$I(\theta) = \left\{ -E \frac{\partial^2 \ln f(x | \theta)}{\partial \theta_i \partial \theta_j} \right\}.$$

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We indicate logarithm to the base two by “log” and the natural logarithm by “ln.” Originally this prior was constructed by invariance arguments for the purpose of capturing the elusive idea of no prior knowledge.

Recently, these studies were carried further by Clarke and Barron [3], [4], who were able to provide a very accurate asymptotic formula for the redundancy of the code, defined by the mixture density

$$f_w(x^n) = \int f(x^n | \theta) dw(\theta)$$

namely

$$E_\theta \ln \frac{f(x^n | \theta)}{f_w(x^n)} = \frac{k}{2} \ln \frac{n}{2\pi e} + \ln \frac{|I(\theta)|^{1/2}}{w(\theta)} + o(1) \quad (2)$$

when the modeled processes are independent and identically distributed (i.i.d.), satisfying suitable smoothness conditions. From this, an asymptotic formula for the channel capacity follows if we take the prior as Jeffreys' prior (1).

Gradually over the years, universal coding has evolved into something that could be called *universal modeling*. The purpose is no longer restricted to just encoding of data but rather to finding optimal models, above all an optimal universal model, to be used whenever models of the data-generating machinery are needed. Universal modeling with the associated Minimum Description Length (MDL) principle for statistical inference, then, generalizes the older idea of parameter estimator in statistics [12], [15], and it incorporates the model complexity which affects all aspects of model performance.

Instead of measuring a code's performance by the mean code length as in universal coding, the central question of interest in universal modeling is the code length achievable for individual sequences. This is because the intent is to judge the goodness of a model or a model class by the shortest code length we can encode the data with, while using the restrictions in the string provided by the model/class, which cannot be done by the mean length. After all, we can construct a model so that a certain easy-to-encode string, such as one with all the symbols equal, will get the probability $1 - \epsilon$, while all the other strings share the remaining ϵ . With such a model, we obtain as short a mean length as we wish, but clearly it is of no use for making inferences.

In an extension of Shannon's information $-\log P(x^n)$ of the sequence x^n , the shortest code length, obtainable somehow with the help of the model class and eventually called the *stochastic complexity*, was taken to represent the information in the sequence, when a class of processes is given rather than just one process, as in Shannon's case. This left the burning

question of how to compute it. Much as Shannon's coding theorem, which states that the length of an optimal prefix code will recapture the data generating distribution, an extended coding theorem was proved [13], [14], to a similar effect: There exist a best code and the process f^* it defines, which asymptotically behaves as whatever process in the family generates the data. More precisely, for almost all parameter values θ

$$\lim_{n \rightarrow \infty} \frac{E_\theta \log [f(x^n | \theta) / f^*(x^n)]}{\frac{k}{2} \log n} = 1 \quad (3)$$

and the left-hand side cannot be made smaller no matter which process is substituted for f^* ; for a recent generalization see [11]. The expectation operation can under suitable conditions be dropped, and the ratio in the left-hand side can be less than unity only for finitely many values of n , almost surely. A number of codes reaching the lower bound can be constructed with the asymptotic length

$$-\log f(x^n | \hat{\theta}) + \frac{k}{2} \log n + o(\log n) \quad (4)$$

where $\hat{\theta}$ is the maximum-likelihood estimate, and this has served as the formula for the stochastic complexity until now.

Unfortunately, the extended coding theorem cannot be sharpened to distinguish between code lengths that differ by a constant, however large, and the second term in the right-hand side of (2) suggests that the constant term can be large indeed, if the Fisher information of the data-generating source is nearly singular. Hence, such a code length as the stochastic complexity would not serve the intended purpose to provide a yardstick by which model classes can be compared in the light of a finite and possibly even small amount of data. For this reason we set out to reconsider the problem of coding data sequences in a nonredundant manner as efficiently as we can while paying attention to any potentially large additional terms that may arise.

Among the known coding procedures one stands out as an intuitively appealing candidate for the sought-for code, the so-called maximum-likelihood (ML) code, defined by

$$\hat{f}(x^n) = \frac{f(x^n | \hat{\theta}(x^n))}{\int f(x^n | \hat{\theta}(x^n)) dx^n} \quad (5)$$

and proposed for finite alphabets in [6] and [19] but without an explicit easy-to-calculate formula for the code length. Clearly, for infinite alphabets the integral must be finite for the code to exist. By studying an implementable version of this code, in which the maximum-likelihood estimates $\hat{\theta}(x^n)$ are quantized, we first show that it is equivalent with a two-part code, as discussed in [12], with the inherent incompleteness removed. Then, as the main result, we derive the following formula for the limiting ideal code length:

$$L^*(x^n) = -\ln f(x^n | \hat{\theta}) + \frac{k}{2} \ln \frac{n}{2\pi} + \ln \int \sqrt{|I(\theta)|} d\theta + o(1). \quad (6)$$

The validity of this formula does not require the model class to consist of i.i.d. processes. Instead, we require only

that the maximum-likelihood estimates satisfy the Central Limit Theorem (CLT) along with certain weak smoothness conditions. In particular, the equality between (6) and the code length of the complete two-part codes studied here sharpen the bounds in [1, Theorem 4.4] while at the same time extending the result beyond the i.i.d. processes.

It seems clear that these properties, above all the expression (6) for the maximum-likelihood code (5), lends this code length a unique status, and we define it to be the *stochastic complexity* of the string x^n , given the model class, thereby concluding a decade-long search. We finish the paper by describing how the requirement of the integrability of Jeffreys' prior can be removed, which we illustrate for the exponential and Gaussian processes. The explicit formula (6) turns out to make the required modifications quite easy. By contrast, the evaluation of Jeffreys' mixture, when the integration is restricted to a subset of the entire space, as the case would be with the Gaussian distributions, could provide difficulties.

II. THE MODEL CLASS

The model class of interest in universal modeling studied in this paper is a class of parametric processes $\mathcal{M}_k = \{f(x^n | \theta)\}$ defined by a density likelihood function, such that the marginality condition

$$\int f(x_1, \dots, x_{n+1} | \theta) dx_{n+1} = f(x_1, \dots, x_n | \theta)$$

is satisfied for all n and for each parameter θ in an open bounded subset Γ of R^k . This set is chosen to include the parameters and their maximum-likelihood estimates of interest. For the construction of a universal model the appropriate class is the union of $\mathcal{M}_k$ over the number of parameters and actually even over different structures, each having the same number of parameters, as the case often is. However, in this paper we concentrate on the smaller class $\mathcal{M}_k$. We take the parameterization to be irreducible so that any two different parameter values define distinct density functions and processes, and we consider the following conditions:

i) For the matrix $I(x^n, \theta)$ of the elements

$$I_{ij}(x^n, \theta) = -n^{-1} \partial^2 \ln f(x^n | \theta) / \partial \theta_i \partial \theta_j$$

$$I_n(\theta) = -n^{-1} \left\{ E \frac{\partial^2 \ln f(x^n | \theta)}{\partial \theta_i \partial \theta_j} \right\} \rightarrow I(\theta) \quad (7)$$

the limit satisfying $0 < c_1 \leq |I(\theta)| \leq c_2 < \infty$ for all $\theta \in \Gamma$.

ii) The elements of $I(\theta)$ are continuous in Γ , and
iii)

$$\int_{\Gamma} \sqrt{|I(\theta)|} d\theta < \infty.$$

iv) Further, the maximum-likelihood estimator satisfies the CLT: The distribution of $\xi = \sqrt{n}(\hat{\theta}(x^n) - \theta)$ converges to the normal distribution with mean zero and covariance $I^{-1}(\theta)$, uniformly for $\theta \in \Gamma$. Hence, in particular, if $R_r(0)$ is a rectangle of side length r centered at

the origin, its probability mass, induced by $f(x^n | \theta)$, satisfies

$$P\{\xi \in R_r(0)\} \rightarrow \frac{|I(\theta)|^{1/2}}{(2\pi)^{k/2}} \int_{R_r(0)} e^{-\frac{1}{2}\|\xi\|_{I(\theta)}^2} d\xi. \quad (8)$$

v) Finally

$$I(x^n, \hat{\theta}) = -n^{-1} \left\{ \frac{\partial^2 \ln f(x^n | \theta)}{\partial \theta_i \partial \theta_j} \right\}_{\theta=\hat{\theta}} < C_0 < \infty \quad (9)$$

where C_0 is a positive-definite matrix and x^n is such that the maximum-likelihood estimate $\hat{\theta} = \hat{\theta}(x^n) \in \Gamma$. In addition, the family

$$I_{ij}(x^n, \theta(\xi)) = -\frac{1}{n} \partial^2 \ln f(x^n | \theta(\xi)) / \partial \xi_i \partial \xi_j$$

for $n \geq 1$, $1 \leq i, j \leq k$, as a function of the normalized variable ξ is equicontinuous at $\xi = 0$, where $\theta(\xi) = \hat{\theta}(x^n) + \xi/\sqrt{n}$.

In order to satisfy these conditions, which may not be independent, the range of parameters may have to be restricted to exclude singular points of the Fisher information $|I(\theta)|$. For instance, if $|I(\theta)|$ becomes infinite at a finite number of points, as the case is with the Bernoulli class, where $I(\theta) = 1/(\theta(1-\theta))$, we cut out a small neighborhood around these points and consider the interior of the remaining set as the range Γ . If Condition iii) holds, we may then let the cut-out neighborhood shrink to zero.

Generally speaking, the CLT holds for other than i.i.d. processes, if the processes defined by the likelihood function, one for each parameter value, are such that $f(x_t | \theta)$ and $f(x_{t-\tau} | \theta)$ become independent as τ goes to infinity. In many important classes of processes, such as the finite alphabet Markov and the ARMA processes, these conditions are indeed satisfied [10].

Notice that the Condition v) is weaker than the equicontinuity of the family of functions $I(x^n, \theta)$ of the unnormalized variable θ in that $|I_{ij}(x^n, \theta) - I_{ij}(x^n, \hat{\theta})| \leq \epsilon$ whenever $\|\xi\| \leq \delta_\epsilon$ or $\|\theta - \hat{\theta}\| \leq \delta_\epsilon/\sqrt{n}$, uniformly in n . Together with (9), it implies uniform boundedness of $I(x^n, \theta)$ in the neighborhood of $\hat{\theta}$ of radius $r/\sqrt{n}$, for $r \leq \delta_\epsilon$. Condition v) is particularly easy to verify for many classes of models where $\hat{\theta}(x^n)$ is a sufficient statistic so that $f(x^n | \theta) = h(\theta, \hat{\theta}, n)g(x^n)$ and for which a formula is given for the first factor.

III. STOCHASTIC COMPLEXITY

Only discrete data can be encoded with a finite code length $L(x^n)$, and any complete prefix code defines a distribution $P(x^n) = 2^{-L(x^n)}$, while the converse is true if we take $L(x^n)$ as the least integer upper bound $\lceil -\log P(x^n) \rceil$, the logarithm being binary. It will be convenient to ignore the integer-value requirement on the code length and to regard $-\log P(x^n)$ itself as an (ideal) code length. We regard even the negative logarithm of a density $-\log f(x^n)$ of a data sequence of real numbers as a code length with the understanding that an integer-valued code length may be obtained by truncating the real numbers to a fine precision δ and replacing the density by the probability $f(x^n)\delta^n$. This abuse of the words "code

length" is convenient and quite harmless since the precision itself is generally of no importance in our discussion and can be dropped.

We shall derive the main formula for the stochastic complexity as the limit of a code length of two equivalent codes, namely, a complete two-part code and a quantized ML code. The idea of encoding data with two-part coding is first to design a code for the data with the length function $-\log f(z^n | \hat{\theta})$, where $\hat{\theta} = \hat{\theta}(x^n)$ is the maximum-likelihood estimate of the parameters, obtained from the observed sequence x^n , and z^n runs through all the data sequences. We then need to add to the code string for the data the parameter value used in encoded form. This preamble must be a prefix code so that the binary encoded parameter value of some length $L(\hat{\theta})$ can be separated from the subsequent binary encoded data without a comma. Clearly, the maximum-likelihood estimate must be quantized to some precision $\hat{\theta}^d = \hat{\theta}^d(x^n)$, which we take as the center of a rectangle $R_d(\hat{\theta}^d)$ of side length d , whose sides are parallel with the coordinate axes in the k -dimensional space. We can then encode the data with the code length

$$-\log f(x^n | \hat{\theta}^d) + L(\hat{\theta}^d). \quad (10)$$

But we can do better, for such a code is incomplete and the codewords needlessly long. To see this, notice that once we have decoded $\hat{\theta}^d$ from the preamble, we know that the subsequently encoded data must be such that the ML estimate $\hat{\theta}(x^n)$ falls within the equivalence class $R_d(\hat{\theta}^d)$ and not outside. If we write

$$P_d(\theta^d) = \int_{\hat{\theta}(x^n) \in R_d(\theta^d)} f(x^n | \theta^d) dx^n \quad (11)$$

we can encode the data, including the parameter estimates, with the length

$$L_d(x^n) = -\log \frac{f(x^n | \hat{\theta}^d)}{P_d(\hat{\theta}^d)} + L(\hat{\theta}^d). \quad (12)$$

This clearly is shorter than the code length (10), which therefore is just a more or less crude estimate of the shortest code length. In fact, we see immediately that (12) is complete in that $2^{-L_d(x^n)}$ integrates to unity, provided $L(\hat{\theta}^d)$ itself is complete. Because of the obvious connection between (10) and the maximized posterior $f(\theta^d | x^n)$ with the prior defined by $L(\theta^d)$, the MDL principle itself has sometimes been erroneously identified with the posterior maximization principle. The difference can be stated clearly in the case of a finite number of models, i.e., hypotheses: The MDL model provides the best explanation of the data in terms of the code length, while the posterior maximizing model is the most likely one. To make the latter meaningful we must make the awkward assumptions that the models themselves are outcomes from a random variable and that one of them is the "true" hypothesis.

The completeness of a code is, of course, a minimum requirement, and we want more. To see why, let $Q(\theta^d) = 2^{-L(\theta^d)}$ be the prior distribution for the quantized parameters defined by the prefix code length, and rewrite (12) as

$$L_d(x^n) = -\log f(x^n | \hat{\theta}^d) + \log \frac{P_d(\hat{\theta}^d)}{Q(\hat{\theta}^d)}. \quad (13)$$

When the Fisher information has singularities, the ratio in the second term can be very large for most priors. The reason is that the probability $P_d(\hat{\theta}^d)$ is proportional to the square root of the Fisher information, as will be shown shortly. What makes this phenomenon particularly unpleasant is that it tends to happen for strings that have strong statistical regularities permitting a short encoding. An example is a binary sequence in which the ratio n_0/n of the number of zeros to the length of the sequence is either small or close to unity, and the model class is the Bernoulli family with the familiar uniform prior. The same is true for the normal family with most conjugate priors, if the empirical variance is small.

The anomaly can be avoided either by adjusting the equivalence classes $R_d(\theta^d)$ so that their volume is inversely proportional to the square root of the Fisher information, or by selecting the unique prior

$$Q(\theta^d) = \frac{P_d(\theta^d)}{\sum_{\eta^d} P_d(\eta^d)} \quad (14)$$

which cancels the trouble-causing probability $P_d(\hat{\theta}^d)$. Here η^d runs through all the quantized parameter values. We opt for the much simpler second method. With (14) we have then

$$L_d(x^n) = -\log f(x^n | \hat{\theta}^d) + \log \sum_{\eta^d} P_d(\eta^d) \quad (15)$$

where only the first term depends on the data. Notice how this prior matching echoes the reasoning in [3] in terms of the mean redundancy of the mixture densities (2). We also see that such a two-part code is equivalent with the quantized ML code, based on the process

$$\hat{f}_d(x^n) = \frac{f(x^n | \hat{\theta}^d)}{\sum_{\eta^d} P_d(\eta^d)}. \quad (16)$$

The limiting case for $d = 0$ of this code

$$\hat{f}(x^n) = \frac{f(x^n | \hat{\theta}(x^n))}{\int_{\hat{\theta}(x^n) \in \Gamma} f(x^n | \hat{\theta}(x^n)) dx^n} \quad (17)$$

which we shall justify presently, was in the case of finite alphabet processes, where the integral is replaced by a sum, proposed in [6] and [19] and called the maximum-likelihood (ML) universal code. As we shall see, the assumptions in Section II imply that the integral is finite.

Clearly, both of the codes defined by (16) and (17) are complete. This has the implication that neither code can be shorter for all strings, and to find a rational basis to prefer one over the other requires other considerations. We can state, however, that in order for a two-part code to reach the lower bound in (3), the precision must increase at least as $d = O(1/\sqrt{n})$. For this reason we regard the limit $-\ln \hat{f}(x^n)$ in (17) as the candidate for stochastic complexity, to be justified by additional properties. Some of these, at least for Markov processes, have already been discussed in [20]. The limiting code length has the additional advantage that both the data and the parameters may be taken as infinite precision numbers, as befits the intended abstract measure of a string's complexity.

The main task in this paper is to derive an asymptotic expression for the code length (15) and to prove that it provides an increasingly accurate estimate of the code length $-\ln \hat{f}(x^n)$ under the stated assumptions, as n grows.

Theorem 1: Let a model class satisfy the Assumptions i)–iv) in Section II. Then, for a sequence $d_n = o(1/\sqrt{n})$, i.e., $d_n = r_n/\sqrt{n}$, where $r_n \rightarrow 0$

$$\ln \sum_{\theta^{d_n}} P_{d_n}(\theta^{d_n}) = \frac{k}{2} \ln \frac{n}{2\pi} + \ln \int_{\Gamma} \sqrt{|I(\theta)|} d\theta + o(1). \quad (18)$$

Further, if Assumption v) also holds, then for sequences x^n such that $\hat{\theta}(x^n) \in \Gamma$

$$-\ln \hat{f}(x^n) = -\ln f(x^n | \hat{\theta}(x^n)) + \frac{k}{2} \ln \frac{n}{2\pi} + \ln \int_{\Gamma} \sqrt{|I(\theta)|} d\theta + o(1). \quad (19)$$

Remarks: Unless Γ is the entire parameter space, we select it so that it includes the maximum-likelihood estimate $\hat{\theta}(x^n)$ and the integral in (19) can be either calculated or estimated. Our proof of (18) does not give a recipe for the rate with which r_n goes to zero, and the term $o(1)$ will have to be estimated by other means when needed. For Bernoulli processes it shrinks rapidly; see Example 1 below.

We define

$$L^*(x^n) = -\ln \hat{f}(x^n) \quad (20)$$

to be the stochastic complexity of the string x^n , relative to the model class considered. It serves as a measure of the complexity of the string, or its information, much as Shannon's information does relative to a single distribution. Strictly speaking, it is not a code length, but it can be well approximated by a real code length designed with the help of the quantized maximum-likelihood estimate. It is a fortunate fact that this expression for the stochastic complexity includes explicitly the density function $f(\cdot | \hat{\theta}(x^n))$, which may be taken as the idealized optimal model. Compare this with the older versions [15], defined in terms of mixtures, which are determined by the models in the class but which themselves do not define any model in the class, except when a special prior makes the mixture equivalent with (19).

The MDL philosophy deals with models of data, i.e., models fitted to the data, which makes them functions of data. All other models, often as various limits of the fitted ones, are mathematical abstractions, which, of course, can be useful on other grounds. Although the entire model class, where the parameters range over a continuum, clearly has infinite complexity in terms of the code length, the models we actually fit will have to have their parameters quantized to a finite precision. For optimality to within the second term in (19), this precision must behave as $O(1/\sqrt{n})$, and in order to capture the third term as well it has to behave as $o(1/\sqrt{n})$. Hence, the number of distinguishable models grows with the length of the data, which seems reasonable. In view of this we define the model complexity (as seen through the data)

$$L_n(\mathcal{M}) = \ln \frac{f(x^n | \hat{\theta}(x^n))}{\hat{f}(x^n)}.$$

Proof: Introduce the change of variables $\hat{\theta} = \theta + \xi/\sqrt{n}$, which transforms the rectangle $R_r(0)$ into $R_{r/\sqrt{n}}(\theta)$. Then the right-hand side of (8), denoted $\bar{P}_d(\theta^d)$, equals the integral of the normal density function with mean θ^d and covariance $n^{-1}I^{-1}(\theta^d)$ over the rectangle $R_{r/\sqrt{n}}(\theta)$. With $P_d(\theta^d)$ denoting the left-hand side of (8), or (11), we have for $d = r/\sqrt{n}$ and θ as the center θ^d of this rectangle

$$|P_d(\theta^d) - \bar{P}_d(\theta^d)| \leq \epsilon \quad (21)$$

which holds for all positive numbers ϵ and $n \geq m_\epsilon$. It will be convenient to write

$$\epsilon = r \left(\frac{r}{\sqrt{2\pi}} \right)^k$$

and the corresponding numbers m_ϵ as m_r . We have the estimate for the second term

$$\begin{aligned} \left(\frac{n}{2\pi} \right)^{k/2} |I(\theta^d)|^{1/2} K(r, \theta^d) \left(\frac{r}{\sqrt{n}} \right)^k &\leq \bar{P}_d(\theta^d) \\ &\leq \left(\frac{n}{2\pi} \right)^{k/2} |I(\theta^d)|^{1/2} \left(\frac{r}{\sqrt{n}} \right)^k \end{aligned} \quad (22)$$

where the ratio $K(r, \theta^d)$ of the minimum and the maximum values of the rescaled normal density function in the rectangle satisfies by the assumed property of $I(\theta)$ in i)

$$e^{-cr^2} \leq K(r, \theta^d) < 1$$

for a positive number c . Next, cover the space Γ of the parameters with nonoverlapping rectangles $R_d(\theta^d)$, $d = r/\sqrt{n}$, and form the sums of the two terms in (21) over the centers of the rectangles within the closure of Γ . This leaves out the rectangles that cut through the boundary, which, however, does not alter the integrals as limits of the sums that we form by letting n grow. From (21) and (22) we get

$$\begin{aligned} e^{-cr^2} \sum_{\theta^d} \left[(|I(\theta^d)|^{1/2} - r) \left(\frac{r}{\sqrt{n}} \right)^k \right] &\leq \sum_{\theta^d} \frac{P_d(\theta^d)}{(n/2\pi)^{k/2}} \\ &\leq \sum_{\theta^d} \left[(|I(\theta^d)|^{1/2} + r) \left(\frac{r}{\sqrt{n}} \right)^k \right] \end{aligned} \quad (23)$$

for all $n > m_r$. The sum

$$\sum_{\theta^d} (r/\sqrt{n})^k$$

differs from the volume of Γ , V_Γ , by at most $\sqrt{k}r/\sqrt{n}$ times the surface area of Γ , which clearly does not depend on r or n . Hence, we get the inequality

$$\sum_{\theta^d} (r/\sqrt{n})^k \leq V_\Gamma + K_1 r/\sqrt{n} \quad (24)$$

for some positive constant K_1 . By ii) for each positive number ϵ' , $|I(\theta)|^{1/2}$ differs from $|I(\theta^d)|^{1/2}$ by less than ϵ' whenever

$\|\theta - \theta^d\| \leq \delta(\epsilon')$ and hence for all $\theta \in R_d(\theta^d)$ for large enough n . Therefore we have with iii) and (24)

$$\begin{aligned} \left| \sum_{\theta^d} |I(\theta^d)|^{1/2} \left(\frac{r}{\sqrt{n}} \right)^k - \int_\Gamma |I(\theta)|^{1/2} d\theta \right| \\ \leq \epsilon' \sum_{\theta^d} \left(\frac{r}{\sqrt{n}} \right)^k \leq \epsilon' (V_\Gamma + K_1 r/\sqrt{n}) \end{aligned}$$

which gives lower and upper bounds for the first sum term. Putting $\epsilon' = r$ and introducing these bounds into (23) we deduce the inequalities

$$-cr^2 - Kr \leq \ln \frac{\sum_{\theta^d} P_d(\theta^d)}{(n/(2\pi))^{k/2} \int_\Gamma |I(\theta)|^{1/2} d\theta} < Kr \quad (25)$$

after taking logarithms, which hold for all $r \leq r_K$ and $n > n_r$, some large enough n_r , and K a positive constant. To see this, let K_2 be a constant greater than $2V_\Gamma / \int_\Gamma |I(\theta)|^{1/2} d\theta$, which in turn exceeds

$$K(n) = 2(V_\Gamma + K_1 r/\sqrt{n}) / \int_\Gamma |I(\theta)|^{1/2} d\theta$$

for all n large enough. Pick $K > K_2$ and $a = K/K_2$. Then

$$-Kr = -aK_2 r \leq \ln(1 - K_2 r)$$

for r so small that $-K_2 r$ is greater than the negative solution to $\ln(1+x) = ax$. From this and the concavity of the logarithm function, the first inequality in (25) follows for all large enough n . The second inequality follows from the inequalities

$$\ln(1 + K(n)r) \leq \ln(1 + Kr) \leq Kr.$$

We complete the proof by linking r with n . Let m be the smallest integer, greater than or equal to $1/r_K$. Let r take the values $1/m, 1/(m+1), \dots$, which gives a sequence of the integers $n_{1/m}, n_{1/(m+1)}, \dots$ required for (25) to hold. Next, define a sequence $\alpha(n)$ as follows: $\alpha(n) = 1/m$ for $n_{1/m} < n \leq n_{1/(m+1)}$, $\alpha(n) = 1/(m+1)$ for $n_{1/(m+1)} < n \leq n_{1/(m+2)}$, $\dots$. With δ_r denoting $Kr + cr^2$ for short we define another sequence $\mu(n) = \delta_{\alpha(n-n_{1/m})}$ for $n > n_{1/m}$. Then, if we put $d = \alpha(n)/\sqrt{n}$

$$\left| \ln \frac{\sum_{\theta^d} P_d(\theta^d)}{(n/(2\pi))^{k/2} \int_\Gamma |I(\theta)|^{1/2} d\theta} \right| \leq \mu(n) \quad (26)$$

holds for these values of n . To make this true for all values of n , define $\mu(n)$ for $n \leq n_{1/m}$ as the maximum of the absolute value of the middle term in (25). This proves (18).

To prove (19) we get by Taylor's formula for $\hat{\theta}$ in the rectangle $R_d(\theta^d)$ for $d = r/\sqrt{n}$ and r small enough

$$\ln \frac{f(x^n | \hat{\theta})}{f(x^n | \theta^d)} \leq Cr^2$$

for some positive constant C . This is true because of (9) and the equicontinuity assumption in v). From (25) and (16) we

then obtain

$$\left| \ln \frac{f(x^n | \hat{\theta})}{\hat{f}_d(x^n)} - \frac{k}{2} \ln \frac{n}{2\pi} - \ln \int_{\Gamma} |I(\theta)|^{1/2} d\theta \right| \leq Kr + (c + C)r^2 \quad (27)$$

for $r \leq r_K$ and $n > n(r)$, some $n(r)$. Since $\hat{f}_d(x^n)$ integrates to unity, we get from (27) for $\epsilon(r) = Kr + (c + C)r^2$

$$e^{-\epsilon(r)} \leq \frac{\int f(x^n | \hat{\theta}(x^n)) dx^n}{\left(\frac{n}{2\pi}\right)^{k/2} \int_{\Gamma} |I(\theta)|^{1/2} d\theta} \leq e^{\epsilon(r)}$$

for $r \leq r_K$ and $n > n(r)$. By taking logarithms and defining a sequence $\alpha(n) \rightarrow 0$ as in the proof of (18), we get with (16) the claim (19).

IV. INDEPENDENT PROCESSES

In [3] the special mixture process $f_{\pi}(x^n) = \int f(x^n | \theta) d\pi(\theta)$, defined by Jeffreys' prior (1), was intensively studied for i.i.d. processes. Among the major results obtained was the sharp formula for the relative entropy

$$E_{\theta} \ln \frac{f(X^n | \theta)}{f_{\pi}(X^n)} = \frac{k}{2} \ln \frac{n}{2\pi e} + \ln \int_{\Gamma} \sqrt{|I(\eta)|} d\eta + o(1). \quad (28)$$

Since the right-hand side is independent of θ it also gives the minmax redundancy, which further was shown to agree with the channel capacity. For such processes we see at once that (28) holds even when f_{π} is replaced by the universal process $\hat{f}(x^n)$ in (17). This follows from (19), because for i.i.d. processes [3]

$$E_{\theta} \ln \frac{f(X^n | \hat{\theta})}{\hat{f}(X^n | \theta)} \rightarrow k/2. \quad (29)$$

Remarks: Returning to (28), we see that it not only gives the minmax relative entropy and the channel capacity, as shown by Clarke and Barron in [3], but f_{π} (and hence $\hat{f}$, too) actually is in the same asymptotic distance from all the models in the class, when the distance is measured in terms of the relative entropy. It therefore sits literally at the center of a hyperball, and the radius, given by this common distance, is the mean model complexity adjusted by the bias $k/2$ in (29). While the radius itself grows with n , the per-symbol radius shrinks to zero, and the per-symbol stochastic complexity approaches the entropy, no matter which process in the family generates the data, implying that the universal process represents in the limit all the constraints in the data. Moreover, by the extended coding theorem, no other process approaches the limit at a faster rate. In the literature of universal coding the per-symbol radius is called the (mean) code redundancy. We regard this as a misnomer, for it clearly suggests that the code length is needlessly long and that it could be shortened by a better design of the code. In reality, this is not the case, unless we correctly guess the imagined data-generating distribution, which by the extended coding theorem is an event of zero measure.

In the Bayesian literature there has been considerable confusion and uncertainty about exactly when Jeffreys' prior does represent the lack of prior knowledge. With the present

arguments there is none. Whether we consider the mixture f_{π} or the code (20) in the two-part code interpretation, it is Jeffreys' prior (1) as the limit of (14) for $d \rightarrow 0$ that is responsible for these density functions' occupying the center of the model class. In addition, instead of being invariant just for translations, as in the original searches for noninformative priors, no transformation of the parameters can alter the center of the model class, which clearly is defined by the models themselves independent of any particular one-to-one parameterization of them.

Example 1: We conclude this section with an example of the class of independent finite-alphabet processes, where the probability of the occurrence of symbol i is θ_i , $i = 0, \dots, m-1$. For such processes, a universal code was suggested by Krichevsky and Trofimov in [9] with which the per-symbol mean code redundancy converges to zero, uniformly in θ . We now explain that code as an instance of (19). The Fisher information is given by

$$|I(\theta)| = \prod_i \theta_i^{-1}. \quad (30)$$

Its square root turns out to be integrable over the compact space of the parameters, defined by the simplex $0 \leq \theta_i \leq 1$, $\sum_i \theta_i = 1$, and the result is given by the Dirichlet's integral

$$\int \prod_i \theta_i^{-1/2} d\theta_i = \frac{\pi^{m/2}}{\Gamma(m/2)}.$$

When this is substituted into (19) with $k = m-1$ we get the corresponding stochastic complexity, and (28) gives its relative entropy, which when divided by n clearly converges to zero uniformly in θ .

It may be of interest to compare the stochastic complexity of a binary string with just one zero, relative to the Bernoulli class, with the code length obtained with the mixture using the uniform prior. Formula (19) gives the result as about $1.5 \ln n$, while the code length in the latter case is much larger, namely

$$\ln \binom{n}{1} + \ln(n+1) \cong 2 \ln n.$$

The difference is due to the gross overestimation of the model cost in the case with the uniform prior. This has a practical implication for the design of decision trees by the MDL principle, which currently is based upon the overestimated model cost. We also calculated the $o(1)$ term in (19) for the Bernoulli class for a few values of n . It decreases monotonically from 0.674 at $n = 0$ to 0.119 at $n = 40$, and the same behavior appears to continue.

Another important fact was shown in the cited reference; namely, that such a universal code admits the simple recursive implementation

$$P^*(x^n) = \prod_{t=0}^{n-1} \frac{n(x_{t+1} | x^t) + 1/2}{t + k/2} \quad (31)$$

where $n(i | x^t)$ denotes the number of occurrences of symbol i in the string x^t . To find such recursive implementations of the universal process, defined by the stochastic complexity, for the more general processes studied in this paper is an important

major undertaking. For Markov chains it has been done in [21]. A way to generalize the same underlying algorithm context for nonlinear ARMA processes and chaotic processes is discussed in [16].

V. RELAXATION OF SUMMABILITY

Some of the most important model classes are such that the square root of the Fisher information is not integrable nor is the parameter space compact. These include the class of normal distributions as well as the exponential distributions arising in failure-time applications. For such cases the asymptotic formula (19) for the stochastic complexity requires a modification. In this concluding section we illustrate how such cases can be handled by calculating the asymptotic expression for the stochastic complexity for the exponential distributions and the all-important Gaussian family, as needed in the regression problems.

Often, $I(\theta)$ has only one eigenvalue that becomes infinite at the origin, which moreover coincides with one component of θ , say the first. This happens in both of the examples discussed below.

Example 2: Consider the family of i.i.d. processes defined by the exponential densities

$$f(x | \theta) = \theta e^{-\theta x}, \quad \theta > 0. \quad (32)$$

The Fisher information is given by $I(\theta) = 1/\theta^2$, the square root of which is not integrable. In failure-time problems [17], one actually needs the induced distribution on the order statistics, but it has the same Fisher information, and the result is the same except for the leading term $-\log f(x^n | \hat{\theta})$. Let a be the smallest integer such that 2^a is not smaller than the maximum of the numbers $1/\hat{\theta}(x^n)$ and $\hat{\theta}(x^n)$, say $\hat{\mu}$, where

$$\hat{\theta}(x^n) = n / \sum_t x_t.$$

Define the set $\hat{\Gamma} = (2^{-a}, 2^a)$, which plays the role of the interior of a compact set in Section II. We selected these boundaries because they can be encoded with a short code length, in fact with about $c + \log^* \hat{\mu}$ bits, where c is about 2.87 and $\log^* x = \log x + \log \log x + \dots$, the dots indicating iterated logarithms as long as they remain positive [8], [12]. We have

$$\int_{\hat{\Gamma}} \theta^{-1} d\theta = 2a$$

and from (19) we get the code length in terms of both binary and natural logarithms as

$$L^*(x^n) = n(1 - \ln \hat{\theta}) + \frac{1}{2} \ln \frac{n}{2\pi} + \log(2a) + \log^* a + c + o(1). \quad (33)$$

Example 3: Consider the family of Gaussian distributions $\mathcal{M}_k = f(y^n | X, \theta, \tau)$ defined by

$$f(y^n | X, \theta, \tau) = \prod_{t=1}^n f(y_t | \mu_t, \tau) \quad (34)$$

where $X = \{x_{t,i}\}$, $t = 1, \dots, n$, $i = 1, \dots, k-1$ denotes the matrix of the regression variables, and each Gaussian factor has the common variance τ while the means μ_t are given by the linear combination of the regression variables, thus

$$\mu_t = \sum_{i=1}^{k-1} \theta_i x_{t,i}.$$

There are then k free parameters in this class, namely, the variance τ and the $k-1$ parameters θ_i . We consider the basic case where the regression variables satisfy the natural condition

$$\Sigma = \frac{1}{n} X' X > 0$$

and we can write the relevant Fisher information as

$$|I(\theta, \tau)| = \frac{1}{2\tau^{k+1}} |\Sigma|. \quad (35)$$

It is singular for $\tau = 0$, and its square root is not integrable over the space $\Gamma = \{\tau, \theta \mid \tau > 0, \theta \in R^{k-1}\}$.

The maximum-likelihood estimates of the parameters are given by

$$\hat{\tau} = \frac{1}{n} \sum_t y_t^2 - \hat{\theta}' \Sigma \hat{\theta}$$

$$\hat{\theta} = \frac{1}{n} \Sigma^{-1} \sum_t y_t \bar{x}_t \quad (36)$$

where $\bar{x}_t$ denotes the column vector of the components $x_{t,1}, \dots, x_{t,k-1}$. Since the two estimates are independent, knowing one gives no help in encoding the other, and we may encode them separately. In order to carve out the required subset $\hat{\Gamma}$ of the parameters, put first $\Gamma = \Gamma_1 + \Gamma_2$, where $\Gamma_1 = \{\tau, \theta \mid 1 \leq \tau\}$, and Γ_2 denotes the rest. It will be convenient to work out the required formulas for suitably defined open-bounded subsets of Γ_1 and Γ_2 separately.

Define s as the least integer such that $2^{2s} \geq \hat{\theta}' \hat{\theta}$. Consider first the case where $\hat{\sigma} = \sqrt{\hat{\tau}}$ is greater than or equal to unity. Defining

$$\hat{\Gamma}_1 = \{\tau, \theta \mid 1 < \tau, \hat{\theta}' \hat{\theta} < 2^{2s}\} \quad (37)$$

we get at once

$$\int_{\hat{\Gamma}_1} \sqrt{|I(\tau, \theta)|} d\tau d\theta = (2|\Sigma|)^{1/2} 2^{s(k-1)} C(k-1)/(k-1) \quad (38)$$

where $C(k-1)$ denotes the volume of a $(k-1)$ -dimensional unit ball. Let then $\hat{\sigma} < 1$. Take r as the least integer such that $2^{-r} < \hat{\sigma}$, and put

$$\hat{\Gamma}_2 = \{\tau, \theta \mid 2^{-2r} < \tau, \hat{\theta}' \hat{\theta} < 2^{2s}\}. \quad (39)$$

This time the required integral over $\hat{\Gamma}_2$ is the expression (38) increased by the factor $2^{r(k-1)} - 1$. From formula (19) we get the code length for the case $\hat{\tau} > 1$ as

$$\begin{aligned} L^*(y^n | \bar{x}^n) &= \frac{n}{2} \ln \frac{2\pi e \hat{\tau}}{\delta^2} + \frac{k}{2} \ln \frac{n}{2\pi} + \frac{1}{2} \log(2|\Sigma|) \\ &\quad + s(k-1) + \log \frac{C(k-1)}{k-1} + 2c \\ &\quad + \log^* r + \log^* s + o(1) \end{aligned} \quad (40)$$

where the data are expressed to the precision δ . For $\hat{\tau} < 1$ this is to be increased by $\log(2^{r(k-1)} - 1) \cong r(k-1)$. Often, the numbers r and s may be taken as constants for all the models compared, in which case (40) simplifies to

$$L^*(y^n | \bar{x}^n) \cong \frac{n}{2} \ln \frac{2\pi e \hat{\tau}}{\delta^2} + \frac{k}{2} \ln \frac{n}{2\pi} + \frac{1}{2} \log(2|\Sigma|) + (s+r)(k-1) + \log \frac{C(k-1)}{k-1}. \quad (41)$$

This applies to both $\hat{\tau} > 1$ and $\hat{\tau} < 1$.

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